Structures and Architecture. REstructure REmaterialize REthink REuse - Rinke & Frier Hvejsel (Eds)
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An anatomy of design principles for lasting architecture in practice

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ABSTRACT: This paper develops an anatomy of design principles as mediators from theory to practice, to guide the design for lasting architectures of low embodied energy in their use-stage. The paper uses a review of three precedent studies on building change with different approaches to formulating design advice, and uses participatory action research to preliminarily test emergent design principles with key stakeholders as well as in design processes with architecture students. With the aim of delivering value in practice, the paper proposes a framework to tie empirically grounded spatial propositions to quantifiable outcomes by combining positivist and interpretive research approaches. Conclusively, the paper outlines a methodology to sequence design principles for their application and further testing in design processes of architectural practice.

1 INTRODUCTION

In Denmark, buildings are predicted to last 200 years on average (Aagaard et al. 2013). Instead of addressing *how long* buildings last, the paper asks *in what form* they do so. The embodied energy (EE) environmental impacts of renovations, modernisations, and transformations in buildings' use-stage (LCA B-phase) tend to be overlooked in policy and research despite their significance, and interdisciplinary approaches to bridge from academia to practice are sought after (Bowyer et al. 2022; Pomponi and Moncaster 2016; Birgisdottir and Madsen 2017). The current literature in the field of use-stage changes and EE largely falls into separate disciplinary categories: quantitative studies in engineering through analysis of large data sets to define and predict lifespans or resilience (e.g. Haugbølle et al. 2021) and qualitative studies in architecture, often based on single-case studies, exploring processes and concepts of change (e.g. Leatherbarrow 2020). This paper examines how quantitative and qualitative approaches may be synthesised for three purposes: 1) academically, for a deeper and interdisciplinary understanding of the topic of building change, 2) to bridge from academia to practice with the formulation of design principles, and 3) a wider goal to strengthen architectural design methods in response to building changes with knowledge and evidence types that a variety of stakeholders would see value in (Giménez et al. 2020). With this approach, the paper grounds the discussion of use-stage EE – often dominated by engineering knowledge and consultancy – in the architectural profession's qualitative methods and knowledge to design for lasting qualities and shifting user needs over time.

Design principles are, in this paper, "used to specify design knowledge in an accessible form" where the practice-oriented term, "design", is *active* and the term "principle" carries theory that *relates* action and outcome, mediating between theory and practice (Gregor et al. 2020, 1–2). Gregor et al. have observed a tendency of lacking consistency and specificity in the formulation of design principles in the field of design science research and in the information systems literature. In architectural research, ways to define design principles in the study of how buildings

DOI: 10.1201/9781003658641-66

change vary too: From high-level qualitative theorising of how spatial qualities make buildings persist (Fannon et al. 2022, 257), to specific and systematic design advice for practitioners on adaptability (Schmidt (III) & Austin 2016, 275, 88) to more informal building process advice (Brand 1995, 198). To achieve lasting architectural qualities with a lowered use-stage EE that meet user needs over time, this paper explores design principles as both a method of analysis and a bridging device from quantitative to qualitative studies and from theory (in)to practice.

2 METHODS AND RESEARCH QUESTION

The paper assesses the research question: *How may design principles bridge architectural research and -practice in the design for lasting architectural quality and lowered use-stage EE?*

A non-comprehensive *literature review* (Groat and Wang 2013) screens literature via JSTOR, Google Scholar, and Danish libraries within the fields of architecture/building science, firstly on "design"/"architectural" + "principle"/"guideline"/"advice" to identify definitions of design principles, secondly on "architecture"/"building" + "change"/"adapt"/"lasting"/"durable"/"resilient"/"persist"/"lifespan"/"design life" to identify key items and strands. Three key theoretical contributions are comparatively analysed to form a precedent study on the formulation of design advice regarding building change. It is accompanied by two tests in practice of precedent design principles of lasting architectural qualities and lowered building use-stage EE (Hentzer Dausgaard forthcoming). The tests use participatory action research (PAR) that follows an iterative loop of *planning/reflection–action–observation* (Cantrell & Walker 1993). PAR test 1 assesses the form, value, and applicability of precedent design principles with a client and architect, where PAR test 2 focuses on the application in design processes with a one-week architecture student workshop.

The analysis contains three parts to define design principles for practice. These parts are conducted chronologically, oscillating between analysis of insights from PAR and from theory:

1. A comparative analysis of how three key literary works study and give design advice on the topic of *building change*, which is discussed with findings from PAR test 1. This discussion identifies a gap on the study of and design advice for building change and clarifies what is needed in the formulation of design principles for building change to answer the paper's research question.
2. A discussion of two theoretical and methodological perspectives to meet these needs: Firstly, how approaches in quantitative and qualitative case study research in the discipline of organisation and management could contribute to bridge this quantitative/qualitative divide in the study of building change (Eisenhardt 1989; Langley 1999). Secondly, how Sekler's three levels of tectonic analysis could be used as a conceptual framework to transfer design ideas into practice, both at general level and a detailed, material, contextually specific one (Sekler 1965).
3. A discussion of the insights from PAR test 2 of the application of precedent design principles in practice, towards a definition of design principles to bridge from theory to practice.

3 PRECEDENTS

3.1 *Precedent literary works*

To develop a state-of-the-art and understand what is needed from design principles to answer the research question, extant approaches to study building change and to formulate design advice for architectural practitioners were screened in a literature search using concepts emerging from the search on "building change" and "lasting architecture". Three significant strands within the literature were identified: *how buildings change* with evolving needs (Brand 1995; Moneo 2017; Perez de Arce 2014), how to increase *building adaptation* (and flexibility) to ensure useability over time at low cost (e.g. Schmidt (III) & Austin 2016; Till & Schneider 2005; Habraken 2021), and how to increase *building persistence*, to maximise use value over time (e.g. Fannon, et al. 2022; Scott 2014; Hassler 2011). For a deeper scrutiny of design advice regarding how buildings *change*, *adapt*, and *persist*, three precedent literary works are reviewed: Brand's *How Buildings Learn: What Happens after They're Built* (1995), Schmidt (III) & Austin's *Adaptable Architecture: Theory and Practice* (2016), and Fannon, Laboy & Wiederspahn's *The Architecture of Persistence: Designing for Future Use* (2022). In the following, findings are summarised (Figure 1):

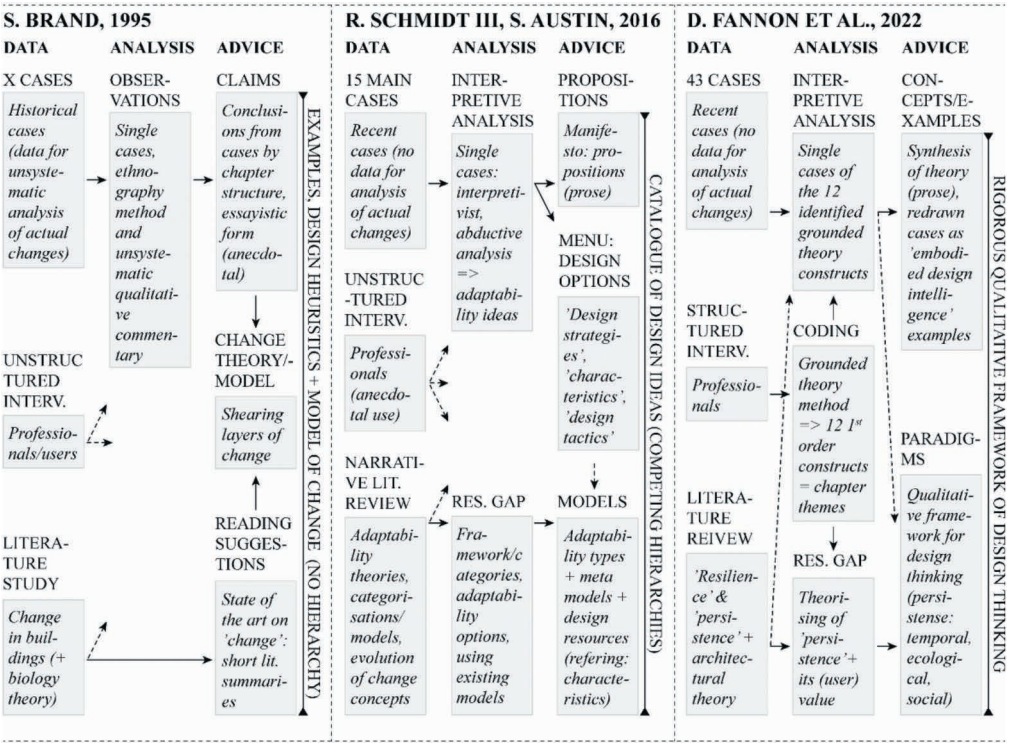


Figure 1. Summary of findings: Review of selected precedent ways to formulate design advice for practice.

Brand's research methodology draws on *ethnography* centring on temporal relationships between users and spaces, mixed with "'qualitative inquiry', 'fieldwork', 'interpretive method', and 'case study'" (Atkinson 2007, 1), without use of a clear case selection strategy. The process of *how* buildings change are synthesised in his model of separate building "layers of change" (Brand 1995, 13) and in design advice of *what* to do, organised into chapters in the form of short prose summaries. There was no clear strategy to conduct or use interviews nor cases to explain the process of change or relationships between spatial variables and outcomes of change.

Schmidt (III) and Austin used literature review on "adaptability" and formed hypotheses of adaptability characteristics from interpretive analysis of 15 recently built single cases to propose specific design advice of *what* to do to increase design-for-adaptability (DfA), presented as options, categories, and frameworks. Interviews with industry professionals on *why* buildings adapt were loosely cited and incorporated into "unravelling contextual contingencies" (Schmidt (III) and Austin 2016, 114) to provide a sense of barriers and opportunities in practice for DfA.

Fannon et al. used grounded theory coding of interviews with industry professionals to identify constructs for systematic, interpretive case study analysis and a literature review focused on "persistence", both to explore *what* makes buildings valuable to keep and thereby persist, and *how* architectural design could facilitate persistence. They kept the richness and complexity of findings without reducing them to actionable design advice, and adapted grounded theory to graphically "code" cases through redrawing which made them further comparable (Fannon et al. 2022, 7).

For the purpose of this paper, two aspects are lacking in the studied approaches: frameworks to rigorously combine quantitative and qualitative approaches, and to offer design advice that has both general applicability as a transferrable idea while keeping spatial, material specificity.

3.2 *Six precedent design principles*

In two preceding pilot case study papers, six design principles were developed (Hentzer Dausgaard et al. forthcoming; Hentzer Dausgaard forthcoming). Based on archival research, the papers featured a longitudinal mapping of every documented change to three public multistorey buildings in Copenhagen over 250 years of use. The design principles were developed by comparing architectural characteristics at the level of components and building layers (Brand 1995) to degrees of change in each case. The degrees of change were quantified with two measures: volume of added material per square meter per year, as well as kg CO_2-equivalents/m^2/year. When a certain spatial variable had an outcome of a significantly lower degree of change in one case compared the others, state-of-the-art analytical categories from the field of engineering, BUILD, Alborg University, Denmark (Aagaard et al. 2013) were used together with an architectural interpretive analysis to explore *why* the spatial variable had the given outcome of change. This analysis resulted in six principles: "ROBUSTNESS HIERARCHY": Increased robustness where most wear is expected. "INCREMENTAL CARE": Incrementally maintainable/repairable spaces and their parts. "LEGIBILITY": Self-evident purposes/value, and upkeep of components for generations. "MULTIPLE PURPOSES": Multiple purposes/qualities within multiple lifespan-parameters of each building part within the wider spatial composition. "SINGULARITY": Low material and compositional complexity of components. "CLEAR BOUNDARY": Some parts to resist expansions and changes through their spatial character. For further details, see: Hentzer Dausgaard forthcoming.

4 RESULTS

4.1 *PAR test 1: Precedent design principles in a project meeting*

To have early feedback from key practitioners – for whom design principles should generate value and be applicable – the six principles were tested in a meeting on the sustainability strategy of the transformation project for a public music hall in Copenhagen. The meeting featured the project architect and a head of sustainability at Realdania, a key philanthropic donor (and co-client) of the project, and the lead architect and partner of LETH & GORI, K. Gori, held on September 18th, 2024. The client valued the design principles as "high-level-design-thinking" architectural strategies, but to have strategic influence as decision making tools in the project, each spatial proposition needed to have its benefits in the project quantified, e.g. in kg CO_2-eq., costs, or m^3 building materials saved p.a. Further, the principles had "too wide margins for interpretation" to credibly lead to the quantified outcomes. Gori commented that the design principles needed "greater spatial specificity or exemplification" and a "method of how to apply them in a design process".

4.2 *PAR test 2: Precedent design principles in design processes*

The second PAR tested the six precedent design principles – this time with insights from both the PAR test 1 and from an emerging analytical approach (see *5.2 Repurposing alternative approaches)* – in a five-day workshop with 3rd-year bachelor students at The Royal Danish Academy, School of Architecture. It was prepared and held by the author, Oct. 9th-11th 2024, and – in summary – asked students to: *use one of the six provided design principles of lasting architecture to reinterpret one of three existing building cases through today's building culture with a focus on either their building envelope, installations, or interior fittings, by developing prototype models in 1:20.* The three existing buildings for reinterpretation were public, multistorey school buildings in Copenhagen: Søkvæsthuset (J.C. Conradi, *1755), Gasværksvejen's School (H.J. Holm, *1880), and Gasværksvejen's School addition (H.C. Hansen, *1971), each undergoing several transformations of use. Key PAR-observations from critics and students in the final jury were:

- *While distinct from each other, design principles have "fuzzy edges" and foster overlap.*
- *From the case study analysis, both patterns of spatial variables resulting in either high or low level of changes should be used to learn from and to develop design principles from.*
- *(How) can spatial variables of low-level change across cases be decoupled from their original physical form and tectonic systems when translated into contemporary building culture?*

5 ANALYSIS

5.1 *Emerging gaps: Learnings from precedent literary works in the light of the PAR test 1*

From the PAR test 1 and precedent literature, three gaps emerge to be addressed in the principles:

1. All three literary studies of building change (3.1) used interpretive research methods with qualitative results but did not include quantitative approaches to measure building change. While Brand has contributed valuable design heuristics for practice (Simmonds & Ing 1998), his findings and design advice lacks a systematic empirical study of *how* and *why* buildings change, beyond illustrative cases. Schmidt (III) & Austin 2016 base their design advice on the intentions/experience of architects to design for adaptability but do not study how buildings *actually* adapt over time. Fannon et al. 2022 could, too, support their theoretical propositions with a longitudinal study. *How could design principles – with the client's comment in the PAR test 1 – relate spatial variables to quantitative outcomes, and thus bridge from quantitative to qualitative analysis?*
2. The precedents operated mainly at a general level of "high-level-design-thinking" (client, PAR test 1) in which ideas about *how* or *why* buildings change, adapt or persist can be transported from one project/context to another. Both Brand and Schmidt (III) & Austin also gave specific design advice of *what* to do to achieve certain outcomes. In Brand's case, specific design advice such as "photographing walls before they close" (Brand 1995, 198) makes sense in the specific illustrative case of balloon-frame housing, yet it would hardly apply to construction types such as brick or concrete walls; the design advice lacks transferability as a design idea. Schmidt (III) & Austin thematically categorised specific design advice within high-level-design-thinking categories, however, their specific design advice was at times neither specific nor general. Advice such as "box-shaped" spaces or "universal image (familiar)" (Schmidt (III) & Austin 2016, 88) is likely too specific to account for variations across e.g. programme needs (e.g. Gibson 2019) yet with "too wide margins" for interpretation to link exact spatial variables to intended outcomes of change (client, PAR test 1). *How could specific design advice credibly transfer design ideas from case studies to building projects while maintaining a direct link to empirical findings?*
3. The form of specific design advice offered in both Brand 1995 and Schmidt (III) & Austin 2016 took the form of a "menu of options" (Schmidt (III) & Austin 2016, 275): "the book is set out to provide a flow of ideas (…) you can dip into" (Schmidt (III) and Austin 2016, 8). Options were in competition or even in contradiction with one another, lacking a method of application. With Gori's PAR test 1 comment, a method to apply design advice or sequence it into a design process is necessary. *How could design principles be rigorously applied in design processes?*

5.2 *Repurposing alternative approaches*

To explore the first question, the paper draws on the field of organisation and management to distinguish between – and eventually combine – different forms of knowledge creation: propositional, positivist studies and qualitative, interpretive studies. On the one hand, positivist studies show *why* variables in a series of compared cases lead to clear, quantifiable outcomes, and can offer propositions of *what* to do, as in *Eisenhardt 1989*. On the other, interpretive studies can identify temporal data patterns with multiple data sources to qualitatively understand *how* processes work, see *Langley 1999*. A third, less widespread but nonetheless rigorous, approach combines positivist and interpretive analyses. Here, the positivist definition of a relationship between variables and outcomes can be expanded on with an interpretive study of the relationship itself and the dynamics of it as a phenomena and process (e.g. in Hotho et al. 2012). Adapted for architectural analysis, this third position could allow design principles to bridge quantitative and qualitative approaches: A positivist approach that studies *why* buildings change and *what* design advice could be given, which establishes a relationship between qualitative spatial variables and quantifiable change-outcomes. And an interpretive approach that studies *how* buildings change, which focuses on the processes that relate variables and outcomes, such as the "ecological", "social" and "temporal" contexts buildings are produced in (Fannon et al. 2022, 283). The interpretive study could include both qualitative and quantitative data, arranged into a timeline of multiple parts from which temporal patterns or trends

could be interpreted with *historical narrative* method (Groat & Wang 2013, 179). In summary, a method to develop design principles is proposed: a positivist analysis that explains *why* and *what* spatial variables impact certain outcomes or patterns of change, followed by an interpretive analysis that qualitatively assesses *how* the relationship and dynamics between spatial variables and patterns of change work, which in turn contextualises and adjusts the positivist explanations and propositions. A design principle, then, is composed of two parts: of "high-level-design-thinking" that explains *why* and *how* a specific aspect of a building changes the way it does, and of "specific design advice" which details *what* the relationship between a spatial variable and a use-stage EE outcome is. Together, these parts synthesise quantitative and qualitative knowledge types to propose, explain, and contextualise design principles.

To address the second question, the paper draws on tectonic theory to propose a precise link between empirical case study findings and the specific design advice, while achieving transferability of the design advice. In the much sited essay "Structure, Construction, and Tectonics"(Sekler 1965), Sekler provides a theory to analyse spaces at three levels: *structure* as the abstractable structural logic or system, *construction* as the technical and material form of the structure, and *tectonics* as the expressive spatial resolution and detailing that relates "form to force" (Sekler 1965, 89). Since Sekler's structure-level operates at a systemic and more conceptual level, design ideas from the empirical case study findings can be transferred to a given design project with a minimum of further interpretation. Sekler's construction and tectonic levels are more contextually specific in terms of materialisation and detailing and would likely need to be reinterpreted to fit the context and the needs of a given design project. The three levels are useful in that they organise the margins for interpretation into three tiers: from little interpretation at the structure-level to more at the others. This, in turn, clarifies how closely aligned a design decision is to an empirical finding (i.e. the pattern in case studies where a spatial variable leads to a desired outcome). Where high-level-design-thinking is textual advice, Sekler's three levels could be used to draw out the specific design advice as spatial constructs at the three levels. Each level could have its own drawing that directly corresponds to the spatial variables observed in the cases. With a constant medium of drawing, a minimum of translation and a higher level of precision and credibility could be achieved, from data collection (archival histories of changes of the selected case study buildings) to analysis (quantitative and qualitative) to the drafting of specific design advice for practice.

5.3 *PAR test 2: An anatomy of design principles for lasting architecture in practice*

To address the third question, the PAR test 2 (see Figure 2) applies precedent design principles in design processes in a student workshop to assess their applicability, potentials, and limitations:

One critic, a professor of architecture, noted: *"While the chosen design principle is most explicit in the prototype, other design principles are present too"*. A student responded: *"the 'edges' of design principles are fuzzy and foster overlaps"*. These comments highlight how design principles are not mutually exclusive but compatible. They can be layered on top of each other as hermeneutical insights (Snodgrass and Coyne 1996), (see Figure 3). Instead of a non-hierarchical menu of competing options, design principles could thus be sequenced into an iterative design process to gradually develop different qualities in a design project. One student group observed in their existing building case that the load bearing façade wall was more "stubborn" and changed less over time than lighter wall types. Therefore, they explored a hierarchy of wall types differentiated by thicknesses and materials in their prototype models so that some walls lend themselves more easily to adaptation and change than others. The students applied the observation of this pattern of change to their prototype proposal "as-found", with only minor adjustments. Other students identified patterns of spatial variables that drove high levels of change in the existing case study buildings, e.g. how load bearing shafts in fired brick were continuously reconfigured in size and form, decade-on-decade. They therefore developed a design response – a "counter option" – to enable shafts to adapt more easily with lower EE over time. Hence, "as-found" spatial variables could be complimented by "counter options" that address design 'shortcomings' found in the case studies. A critic, a professor of architecture, agreed: *"Design principles should equally learn from what existing buildings do well* and *less well over time"*. As students reinterpreted one-meter-thick fired brick walls through today's



Figure 2. Workshop jury room, 11th of Oct., 2024: Models and photos of prototypes of lasting architectures.

building culture, turning towards biogenic materials, a professor of architecture, asked: *"Can we dissociate spatial variables of lasting architecture from an original tectonic materialisation, or is a spatial variable and its tectonic form intrinsically linked?"*. As-found design principles likely need reinterpretation in the context of today's building culture, yet this could happen with the identification of Sekler's levels of structure, construction and tectonics where the structure-level design idea may still be directly transferred into today's practice. Counter options, on the other hand, rely on architects' design knowledge and ability to develop sensible responses to patterns of high levels of change.

In the wake of workshop observations and findings of this paper, the systematic identification of design principles and testing in professionals practice projects can proceed in new PAR-iterations. In the last section, the findings across all parts of the analysis are summarised in Fig. 3.

5.4 *Model: An anatomy of design principles for lasting architecture in practice*

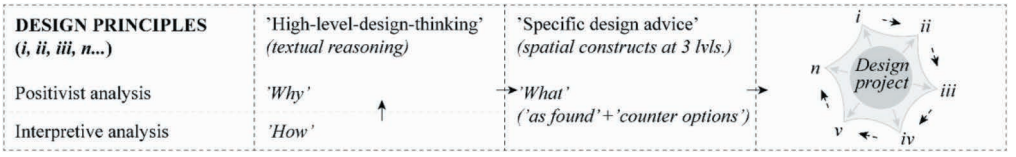


Figure 3. Proposal for an anatomy of design principles for a lasting architecture from theory into practice.

6 CONCLUSION

A research gap on design principles to guide the design of buildings towards lasting architectural qualities and lowered EE of the use-stage was identified. An anatomy of design principles for lasting architecture was developed from both a literature review and comparative analysis of existing studies on design principles regarding building change, PAR tests in practice, and theoretical perspectives. To formulate design principles – both as a method of analysis and as a tool for application in practice – four shortcomings in the precedent literature and precedent design principles were addressed: 1) To combine qualitative and quantitative knowledge types, the design principles integrated a positivist study to explain relationships between spatial variables and quantified outcomes of change with an interpretive study to explain how buildings change, adopted from the field of organisation and management. 2) To clarify and define the relationship between "high-level-design-thinking" and "specific design advice", the design principles devised the above analytical method to link the two, yet with differences in their format of application in practice: textual reasoning of how and why a certain spatial variable shows a tendency towards a certain outcome of change as a framework of high-level-thinking, and drawn constructs from case study findings to propose specific design advice. 3) To credibly transfer design ideas from case studies to building projects while maintaining a direct link to empirical findings, the specific design advice used

Sekler's three levels of tectonic analysis to clarify to what extends designerly interpretation is needed, and how closely aligned a design decision is to the empirical findings.

4) To make design principles applicable in design processes in practice in a rigorous way, a sequencing method was developed to hermeneutically move from one design principle to the next. Future studies could test the design principles developed in this article as a method of analysis and as a tool for application in practice. In turn, tests could refine their formulation and usefulness.

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